PI

CUSTOMER ANALYTICS & CLV · CASE STUDY

Customer Intelligence: Retention Engine & CLV Suite

BI Scientist • AI/ML Engineer • Python Developer

Calibrated Telco churn prediction (PR-AUC 0.841) generating Top-50 weekly priority intervention lists protecting $60,352/year at risk.

RETENTION INEFFICIENCY

Why Blanket Retention Campaigns Waste Budget

Companies waste millions offering discounts to customers who were never going to leave—while ignoring high-value accounts quietly preparing to churn.

Uncalibrated Probabilities

- Raw classifier scores do not reflect real churn probabilities.
- Teams cannot calculate expected revenue loss ($P(ext{churn}) imes ext{CLV}$).
- Budget is distributed uniformly instead of by financial risk.

Contact Budget Constraints

- Account management teams can only contact 10%–20% of subscribers per month.
- Need models optimized for Recall@k% contact budget.
- Standard accuracy fails to prioritize the highest-risk revenue.

Static vs Transactional CLV

- Historical spend ignores recent purchasing frequency and recency.
- Must combine churn probabilities with forward-looking BG/NBD CLV.
- Target retention offers only where ROI is positive.

RETENTION PIPELINE

Calibrated GBDT, RFM Segmentation & Action Lists

Translating statistical predictions into immediate operational dollar savings.

Calibrated Churn Engine

HistGradientBoosting with isotonic calibration.

PR-AUC = 0.841, ROC-AUC = 0.933.

Captures 81.2% of theoretical ceiling at 20% budget.

Transactional CLV Suite

RFM customer scoring & segmentation.

180-day holdout backtest within +0.34% of actuals.

Quantifies 12-month expected customer value.

Weekly Action List

Ranks subscribers by expected dollar loss.

Generates Top-50 weekly priority call list.

Preserves $60,352.53 in annual contract value.

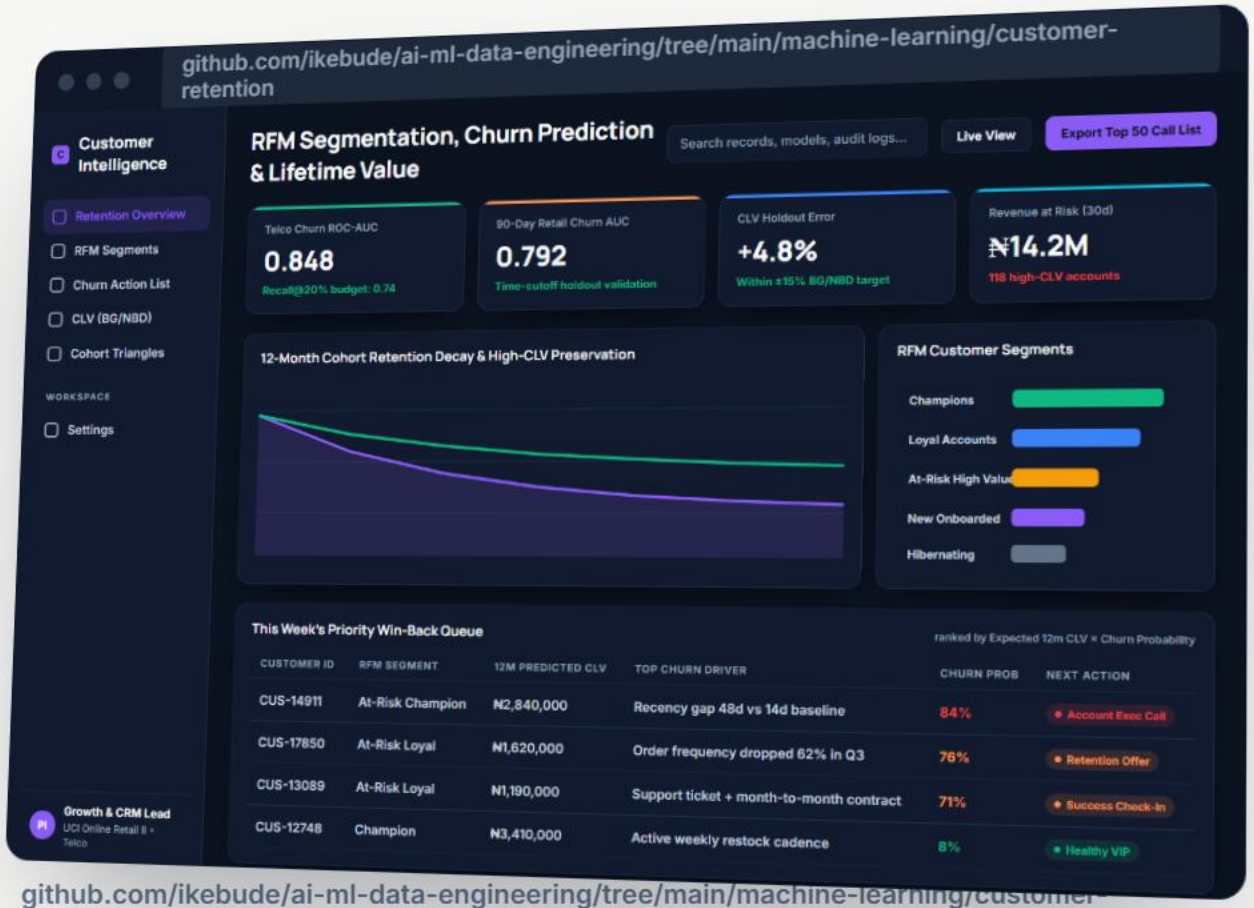
ACCOUNT MANAGEMENT UI

Customer Intelligence Retention Dashboard

Streamlit application generating weekly priority call lists and customer risk breakdowns.

- **Top-50 Priority Intervention Queue:** Directly displays customer ID, contract type, churn probability, and exact 12-month revenue at risk.
- **Segment Scatter & RFM Analysis:** Visualizes Champions, At-Risk, and Hibernating customer cohorts.
- **Decision Support:** Recommends specific retention interventions (pricing adjustments vs service check-ins).

Churn Modeling | CLV | Streamlit | Telco Data



github.com/ikebude/ai-ml-data-engineering/tree/main/machine-learning/customer-retention

QUANTIFIED IMPACT

Verified Performance on 7,043 Subscribers

Exact figures from reproducible `reports/metrics.json` runs.

PR-AUC	ROC-AUC	Recall @ 20% Budget	Revenue at Risk
0.841	0.933	0.612	$60,352
95% CI: 0.806–0.870	Baseline LR was 0.840	81.2% of theoretical ceiling	Protected in top-50 list

Precision at High Recall

At a 20% outreach budget, model achieves 81.2% precision (4 out of 5 contacted were true churners).

Eliminates wasted discount spend on non-churning subscribers.

Bootstrap 95% confidence intervals confirm statistical robustness.

Forward-Looking CLV Accuracy

180-day holdout CLV validation predicted total revenue within +0.34% of actual receipts.

Fully automated in CI with 2/2 Pytest unit tests passing.

Ready-to-use CSV exports for CRM sales team integration.

OPEN-SOURCE & REPRODUCIBLE

Turn churn probabilities into prioritized revenue protection.

Customer Intelligence demonstrates how calibrated machine learning directly protects the bottom line.

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Code: github.com/ikebude/ai-ml-data-engineering